Behavioral Customer Segmentation and Future Purchase Prediction from E-Commerce Event Data

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1. Introduction

1.1 Problem, Motivation, and Impact

E-commerce platforms accumulate large volumes of customer interaction data, including product views, cart additions, cart removals and purchases. Analyzing these interactions may help businesses identify different shopping patterns and predict future purchasing behavior. This project addresses two related problems: identifying behavioral segments among purchasers and predicting whether they will make a repeat purchase within 30 days of their first observed purchase.

Customers follow different paths before purchasing. Some browse many products, while others add items to their carts more frequently. We are interested in understanding these differences and whether they are associated with subsequent purchasing behavior.

The results have practical use for customer retention and marketing. Businesses can use predicted repeat-purchase probabilities to identify customer groups for future promotional experiments, including tests of whether coupons or discounts encourage additional purchases.

1.2 Supervised and Unsupervised Approaches

We use Logistic Regression, Gradient Boosting, and Random Forest in supervised learning to predict repeat purchasing. The models use features related to shopping activity, product exploration, and browsing and cart interactions. We evaluate their performance using ROC-AUC and average precision.

For customer segmentation, we apply K-Means and Gaussian Mixture Models (GMM) to purchasers' pre-purchase behavior. The selected features include interaction proportions, products per session, and browsing breadth. We compare the clustering results using silhouette scores, K-Means inertia, GMM BIC, stability checks, and cluster profiles.

Our results show that K-means best separated purchasers into two distinct customer segments: a cart-oriented group and a view-oriented group. For supervised learning, the best-performing Gradient Boosting model achieved an average precision of 0.1878 and a ROC-AUC of 0.6312 on the temporal test set. These findings suggest that pre-purchase behavior provides some predictive information about repeat purchases, although the model's ability to distinguish repeat purchasers remains limited.

2. Related Work

Using the REES46 cosmetics dataset, Roychowdhury et al. (2021) developed OPAM to predict purchases at the session and user-journey levels. They used **K-Means clustering** for customer segments and **semi-supervised learning** to extend segments to unlabeled events.

Zhang and Wang (2021) focused on whether customers would purchase the same products again, using Alibaba e-commerce data with an improved **Deep Forest** model. Their feature set included user characteristics, product information, and user-product interactions.

Customer segmentation has also been studied using transaction data. Anitha and Patil (2022) applied **K-Means** to Recency, Frequency, and Monetary (RFM) features and used silhouette analysis to evaluate the clustering results.

Our project examines repeat purchasing **at the purchaser level**. We use pre-purchase behaviors to identify behavioral segments and predict whether they make another purchase within 30 days. Two design choices apply throughout: every purchaser in the sample has a complete 30-day follow-up period, and predictive performance is evaluated using a temporal train/test split rather than a random split.

3. Data Sources

3.1 Dataset Source and Format

We use the **eCommerce Events History in Cosmetics Shop** dataset, published by REES46 and available on Kaggle. The dataset contains customer interaction records from an online cosmetics store.

The data are provided as monthly CSV files. Each record contains information about the event timestamp, event type, product, and user.

3.2 Variables, Records, and Time Coverage

The dataset spans from **October 2019 to February 2020** and contains approximately **20 million customer interaction records**. The main variables used in this project are **event_time, event_type, product_id, category, brand, price, user_id,** and **user_session.** The **event_type** variable identifies four types of interactions: **view, cart, remove_from_cart,** and **purchase**.

Multiple events may belong to the same user or shopping session. Exploratory analysis identified **1,639,358 unique users**, of whom **110,518 (6.74%)** had at least one recorded purchase. We focus on eligible purchasers and aggregate their event-level records into a user-level dataset for repeat-purchase prediction and customer segmentation.

4. Feature Engineering

4.1 Data Cleaning and Preprocessing

We combined the monthly event files into a single dataset. Since a user's first recorded purchase may not represent their first-ever purchase, we use each user's earliest recorded purchase as their first observed purchase.

We identified distinct purchase sessions using **user_session**, treating multiple purchase events within the same session as a single purchase occasion. A purchaser was classified as a repeat purchaser if a second distinct purchase session occurred within **30 days** of the first observed purchase.

Then, we applied two eligibility criteria to construct the analysis sample. First, we excluded purchasers whose first observed purchase occurred within the final 30 days of the dataset to ensure a complete follow-up period, leaving **91,979 eligible purchasers**. Second, we excluded purchasers without any recorded interactions before their first observed purchase, ensuring that behavioral features were constructed using only pre-purchase information. This resulted in **91,481 purchasers**, including **12,556 (13.73%)** who made a repeat purchase within 30 days.

We also checked the data quality, including duplicate user IDs and missing values. For customer segmentation, we additionally excluded **21 purchasers** without recorded pre-purchase sessions, leaving **91,460 users** for clustering.

4.2 Behavioral Feature Construction

We aggregated pre-purchase interactions into user-level behavioral features, with **one observation per purchaser.** Events at or after that purchase were excluded to avoid information leakage.

The features cover three categories: **shopping activity** (pre_events, pre_sessions, pre_active_days, pre_unique_products, and pre_duration_days); **interaction counts** (pre_views, pre_carts, and pre_removes); and

behavioral composition (events_per_session, products_per_session, views_per_product, breadth, view_share, cart_share, and remove_share).

The binary target variable, **repeat_30d**, indicates whether a purchaser made a second distinct purchase session within 30 days.

A complete list of the 26 features used in supervised learning, including their definitions and feature-set membership, is provided in Table B10 in the Appendix.

5. Part A — Unsupervised Learning

5.1 Methods Description

5.1.1 Workflow and Feature Representation

Our unsupervised analysis examines whether purchasers can be grouped into interpretable segments based on their behavior before their first purchase. We would also like to experiment whether a derived cluster label provides incremental predictive value in the supervised repeat-purchase task.

We started with 91,481 purchasers with observable pre-purchase activity and excluded 21 without recorded pre-purchase sessions, leaving 91,460 users. We split users chronologically: the earlier 73,168 (80%) formed the training set, and the later 18,292 (20%) formed the test set. The main scaler and clustering models are fitted on training users; the fitted transformations and model are then used to assign test users. This mirrors the intended use of existing behavioral patterns to characterize later purchasers and prevents the test observations from influencing the initial clustering fit.

We chose five behavioral composition features to describe shopping patterns rather than just raw activity: **view_share, cart_share, products_per_session, views_per_product, and breadth**. View_share and cart_share measure the proportions of pre-purchase events spent viewing products and adding them to carts. Products_per_session measures product variety within a session, views_per_product measures how often products are viewed on average, and breadth compares the number of distinct products with the number of events. We did not use repeat_30d to form clusters; we used it later only to describe the groups and in the supervised prediction task.

We dropped remove_share to eliminate perfect multicollinearity, as view_share, cart_share, and remove_share form a closed compositional set that sums to 1, creating a deterministic linear dependency among the features. We also removed events_per_session because it was highly correlated with products_per_session ($r \approx 0.92$). Additionally, we found that Products_per_session and views_per_product were right-skewed, so we applied log1p transformation to both. As a result, their skewness fell from 7.44 to 0.51 and from 3.68 to 0.40, respectively. Finally, we standardized the five features using a StandardScaler fitted on the training set, so that no feature dominates the distance computation through its scale alone.

5.1.2 Methods and Justification

Based on our dataset characteristics and project objectives, we selected **K-Means** and **Gaussian Mixture Model (GMM)** as a starting point. Our dataset contains more than 90,000 purchasers represented by five numerical behavioral features. We needed methods that could handle a large number of users, produce interpretable customer segments, and generate cluster assignments for downstream supervised learning. While other approaches, such as hierarchical or spectral clustering, could also be considered, their computational requirements made them less practical as starting points for our dataset. K-Means and GMM also provide two different ways of modeling customer behavior, allowing us to examine how sensitive the results are to different clustering assumptions: K-Means assigns each user to the nearest cluster center and gives a single label, while GMM models the data as a mixture of Gaussian distributions and provides posterior membership probabilities rather than only hard

assignments. Comparing these approaches helps us understand whether the clustering results are consistent across different modeling assumptions.

K-Means was selected as our baseline because it is computationally efficient and relatively easy to interpret. It assigns each user to the nearest cluster center, making it suitable for grouping users based on similarities in their shopping behavior. Since our clustering features are numerical and standardized, Euclidean distance provides a reasonable basis for comparing users based on their shopping behavior. K-Means also produces cluster centers that can be examined to understand how the groups differ in features. This makes the results easier to interpret and potentially useful for customer segmentation.

GMM was selected to explore whether a more flexible clustering model could capture behavioral patterns that K-Means might overlook. Unlike K-Means, GMM models each group as a probability distribution. We used full covariance matrices so that the model could account for relationships between behavioral features and allow clusters to have different shapes. GMM also provides membership probabilities, which are useful for examining users whose behavior does not clearly fit into one group. This is relevant to our project because shopping behaviors may overlap rather than form completely separate groups.

By comparing K-Means and GMM, we aimed to evaluate both the interpretability of the customer segments and how sensitive the results were to the choice of clustering method. This comparison also allowed us to consider the trade-off between model complexity, cluster separation, and interpretability when selecting the final clustering model.

5.1.3 Hyperparameter Exploration

For both methods, we tested **k values** from 2 to 8. For K-Means, we recorded silhouette scores and inertia, using 10 initializations for each candidate and 20 for the final model. For GMM, we recorded silhouette scores and Bayesian Information Criterion (BIC), using 5 **initializations** during the comparison and 10 for the final model. To assess robustness, we evaluated K-Means stability across 5 **random seeds**, compared cluster alignments using the Adjusted Rand Index (ARI), evaluated random versus temporal splits, and refitted the model on all purchasers to ensure the segments are stable across the full dataset.

5.2 Unsupervised Evaluation

5.2.1 Evaluation Metrics

Silhouette scores were primarily used to measure cluster separation, with higher values indicating greater intra-cluster cohesion relative to inter-cluster distance. For K-Means, we also assessed **inertia**; however, because inertia monotonically decreases with k, an ambiguous elbow cannot independently determine k. For GMM, **BIC** balanced model fitness against model complexity, with lower values preferred. Because BIC and silhouette scores occasionally diverged, we evaluated both metrics alongside their trade-offs rather than treating any single measure as definitive.

We also used the **Adjusted Rand Index (ARI)** to measure how consistently the same users were grouped across different clustering results. An ARI of 1 indicates identical cluster assignments, while a value near 0 indicates agreement close to what would be expected by chance. ARI was particularly useful for our sensitivity analysis, as it allowed us to compare K-Means results across different random seeds, assess agreement between K-Means and GMM, and examine whether the clusters remained consistent when the models were fitted on different sample sizes.

5.2.2 Selection of the Number of Clusters

Both methods achieved their highest silhouette score at k = 2: 0.3191 for K-Means and 0.3033 for GMM. The K-Means inertia curve declined steadily without a clear elbow. However, GMM's BIC continued to decrease as k increased, from approximately 699,713 at k=2 to 262,657 at k=8, indicating that models with more components provided a better fit to the data even after accounting for model complexity. This result conflicts with the silhouette scores, as BIC favors models with more components while silhouette favors k=2.

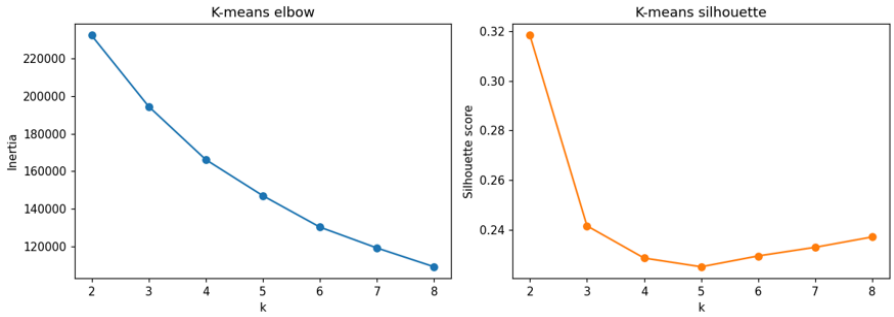


Figure A1. K-Means inertia and silhouette across k ∈ [2, 8].

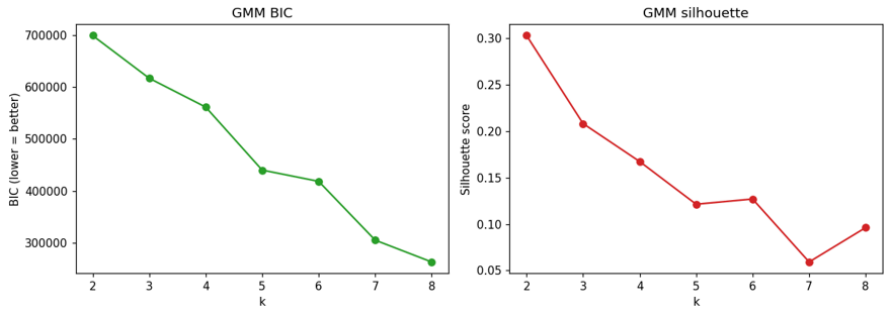


Figure A2. GMM BIC and silhouette across k ∈ [2, 8].

This disagreement is partly explained by what the two metrics measure. Silhouette evaluates how compact and separated the resulting clusters are, whereas BIC evaluates how well GMM fits the data while penalizing model complexity. A more complex distribution may require several Gaussian components to achieve a better fit, even when those components do not represent clearly separated customer groups. Therefore, although the decreasing BIC provides evidence in favor of more components, it does not necessarily mean that a larger number of clusters would be more useful for behavioral segmentation. Therefore, we did not treat BIC as a deciding factor on its own.

To better understand this disagreement between BIC and silhouette, we compared GMM models with two and eight components. The eight-component model had a substantially lower BIC, indicating a better fit to the data after accounting for model complexity. However, its silhouette score was only 0.096, compared with 0.303 for the two-component model. This suggests that the additional components improved the distributional fit without producing more clearly separated customer groups.

We also examined the component centers for k=8 and found more detailed behavioral patterns. These differences suggest that the eight-component model captures behavioral variation that may be lost when using only two groups. However, as shown in Table A1 (Appendix), some component means were also relatively close to each other, such as C0–C5 (1.25), C1–C5 (1.26), and C5–C6 (1.29), compared with much larger standardized Euclidean distances between C2–C3 (5.14) and C2–C4 (4.45). This suggests that some additional components may represent similar behavioral patterns rather than clearly distinct customer segments. We therefore selected k=2 because it provided the highest silhouette scores for both methods and a simpler way to summarize the main differences in customer shopping behavior. However, we acknowledge that k=2 is not the only appropriate choice and may overlook some finer behavioral differences.

5.2.3 Model Comparison and Final Selection

After selecting k=2, we compared K-Means and GMM to determine which method was more suitable for our final customer segmentation. K-Means achieved slightly higher silhouette scores on both the training set (0.3191 vs.

0.3033) and the test set (0.3253 vs. 0.3139). Although the differences were small, the consistent results suggest that K-Means produced somewhat better-separated groups across both datasets.

Table A1. K-Means vs. GMM Metric Comparison (k=2)

Metric	K-Means (k=2)	GMM (k=2)
Train silhouette	0.3191	0.3033
Test silhouette	0.3253	0.3139
Mean pairwise ARI across seeds	0.9986	0.9998

GMM also provided additional information through its membership probabilities. The average maximum membership probability was 0.947, with 83.2% of users receiving a probability above 0.90 and 6.4% below 0.70. This suggests that GMM assigned most users confidently, while a smaller proportion had less certain memberships. Despite the high membership probabilities, GMM's silhouette score remained modest, showing that confident assignments under the fitted probability model do not necessarily correspond to well-separated customer groups. Ultimately, these probabilities reflect the model's internal density estimates rather than empirical ground-truth accuracy.

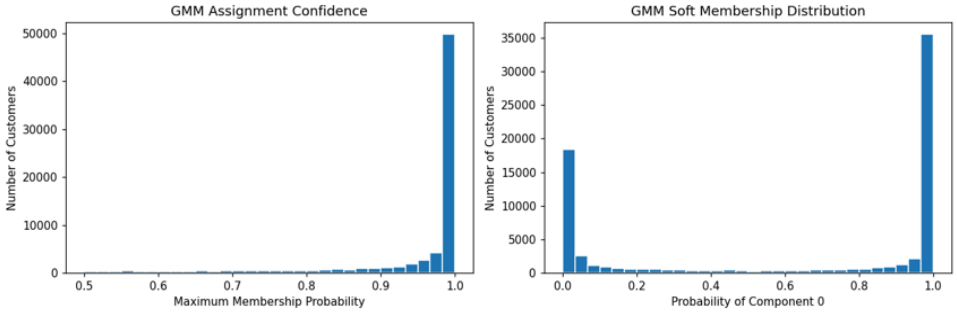


Figure A3. Distribution of maximum posterior membership probabilities for the two-component GMM.

We also compared the stability of both methods across five random seeds using ARI. Both models were highly stable, with mean pairwise ARI scores of 0.9986 for K-Means and 0.9998 for GMM. The cross-model ARI was 0.701, indicating that the two methods produced consistent groupings despite their different assumptions: K-Means relies on distances to cluster centers, while GMM models customers using Gaussian probability distributions. These results suggest that both methods were reliable under the tested initializations, so stability alone did not favor either model.

We ultimately selected K-Means because it consistently achieved slightly better cluster separation on both datasets and provided a direct way to interpret the resulting customer groups, which aligns with our goal of identifying understandable customer segments. However, the performance differences were small, and GMM performed similarly in terms of stability while offering additional probabilistic information. Therefore, our decision reflects the practical needs of this study rather than clear evidence that K-Means is universally superior.

5.2.4 Customer Segment Interpretation

After selecting k=2, we examined the resulting groups to understand which shopping behaviors distinguished them. The standardized feature profiles show clear differences between the two clusters. Cluster 0 (31,832 training users, 43.5%) has higher cart_share (0.754 vs. -0.580), products_per_session (0.353 vs. -0.272), and breadth (0.354 vs. -0.273), suggesting a stronger tendency toward cart activity and broader product engagement. In contrast, Cluster 1 (41,336 users, 56.5%) has higher view_share (0.669 vs. -0.869) and views_per_product (0.688 vs. -0.894),

indicating a stronger tendency on browsing and repeatedly viewing products. Based on these dominant differences, we can roughly describe Cluster 0 as **cart-oriented** and Cluster 1 as **view-oriented**.

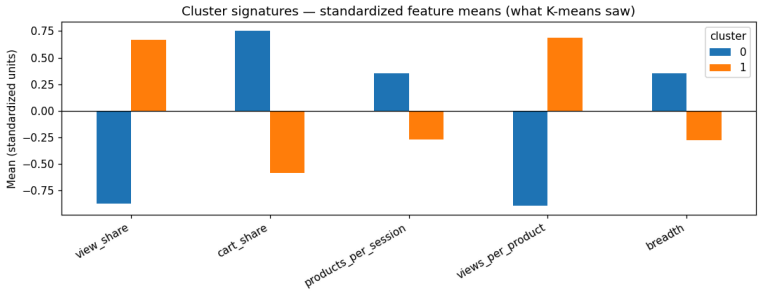


Figure A4. Mean standardized behavioral features for the two final K-Means clusters (training users).

We compared repeat-purchase outcomes and activity across the two clusters (Figure A5). The 30-day repeat-purchase rate was 14.9% for Cluster 0 and 13.4% for Cluster 1, a difference of about 1.5 percentage points. Mean pre-purchase events were approximately 43.0 and 54.7, respectively. Although the difference in repeat-purchase rates was relatively small, it indicates some variation in purchasing outcomes between the two behavioral segments. However, this descriptive comparison alone cannot determine whether cluster membership provides additional predictive value beyond the original behavioral features. We therefore decided to proceed with our original plan to include cluster membership as a feature in supervised learning and tested whether it could improve prediction beyond the original behavioral features.

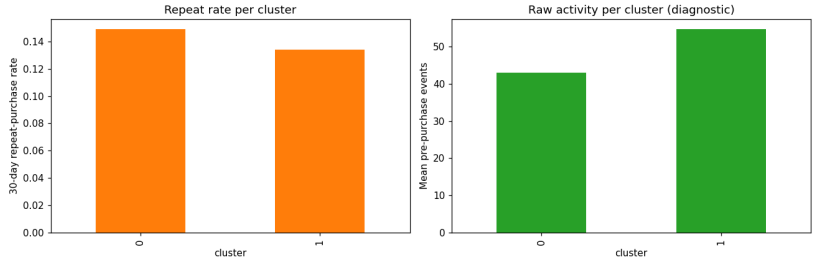


Figure A5. Descriptive 30-day repeat-purchase rates and mean pre-purchase events by K-Means cluster.

5.2.5 Sensitivity Analysis

Random Initialization Stability. We tested whether the K-Means solution changed substantially with random initialization. Across five random seeds at k=2, the mean pairwise ARI was 0.9986. This indicates that almost all users were assigned to the same groups across different runs. Since K-Means can converge to different solutions depending on its initial centroids, these results suggest that our two-cluster solution is not significantly affected by random initialization under the settings tested.

Data Split Sensitivity. We also examined whether the clustering results changed when using different data splits. The K-Means silhouette score at k=2 was 0.320 under the random split and 0.3191 under the temporal split, with k=2 achieving the highest silhouette score among the tested values in both cases. The small difference suggests that the overall cluster separation was not significantly affected by the choice of split.

Sample Sensitivity. To further examine the effect of sample selection, we refitted K-Means using all 91,460 eligible purchasers and compared its assignments with those from the training-only model for the same training users. The ARI was 0.995, indicating that including the additional users had little effect on the original group assignments. Together, these results suggest that the broad behavioral patterns identified by K-Means remained consistent across the tested data splits and fitting samples.

Model Choice Sensitivity. As discussed in Section 5.2.3, we also compared K-Means and GMM assignments at k=2, achieving a cross-model ARI of 0.701. This suggests that the two methods identified broadly similar behavioral patterns. However, the differences in individual assignments indicate that cluster membership is still somewhat sensitive to the choice of clustering model. Therefore, the resulting segments should be interpreted as general behavioral patterns rather than clearly defined customer categories.

Overall, the sensitivity checks show that the K-Means produced consistent results across different initializations, data splits, and samples. However, the modest K-Means silhouette score (0.318) suggests that the two clusters could only represent general behavioral patterns rather than clearly separated customer types. Despite this limitation, we believed it was still worth testing whether cluster labels could improve repeat-purchase prediction besides the original behavioral features.

To proceed, we applied the final K-Means model to all users to generate cluster labels for supervised learning. We also added a split column to indicate whether each user belongs to the training or test set, so the supervised learning stage could maintain the same data split and help prevent data leakage.

6. Part B — Supervised Learning

6.1 Methods Description

6.1.1 Workflow and Feature Representation

The goal of supervised learning is to predict whether a customer will make another purchase within 30 days after their first observed purchase.
We use a user-level dataset with 91,460 customers, including approximately 13.7% repeat purchasers. Each customer has one record containing pre-purchase behavioral features and the binary target repeat_30d.
We use the temporal train-test split from the unsupervised learning section. Customers are ordered by their first observed purchase date, with the earlier 80% used for training (73,168 users) and the later 20% for testing (18,292 users). This helps assess performance on later customers.

Table B1. Feature representations

Set	Feature representation	Features
A	Basic pre-purchase activity: events, sessions, active days, duration	4
B	A plus product browsing, cart activity, and behavior ratios	11
C	B plus cluster_label from unsupervised learning	12
D	C plus 14 additional engineered behavioral features	26

The cluster labels come from a KMeans model with two clusters. The cluster model is fitted on its training portion, and the fitted model is used to assign labels to the test users. Set C tests whether customer segmentation adds predictive value; Set D also includes activity intensity, cart ratios, and activity frequency.

We use information available before the prediction point. The user_id, split, and target columns are excluded from predictors. Median imputation is included in each model pipeline; Logistic Regression also uses feature scaling.

6.1.2 Model Families and Justification

Logistic Regression: A linear classifier used as an interpretable benchmark for associations between customer behavior and repeat purchases.
Gradient Boosting Trees (GBT): Sequentially builds decision trees to capture nonlinear relationships and interactions between behavioral features.

Random Forest: Averages predictions from multiple randomized decision trees, offering a different ensemble method and reducing variance.
We compare these three different model families using feature sets A, B, and C. All models are evaluated using the same five stratified cross-validation folds on the training set. We select the model with the highest mean Average Precision (AP) for further experiments.

6.1.3 Hyperparameter Exploration

GBT-C performs best in the initial comparison. We then test feature engineering (GBT-D), hyperparameter tuning, and balanced class weights. The feature-engineering experiment increases mean AP from 0.2192 to 0.2198.
For tuning, we use RandomizedSearchCV with eight sampled parameter combinations and five-fold CV. The explored parameters are the number of trees, learning rate, maximum tree depth, minimum samples per leaf, and subsample fraction. The selected configuration uses 100 estimators, a learning rate of 0.10, maximum depth of 2, minimum samples per leaf of 5, and subsample fraction of 0.8.

Balanced sample weights are computed separately within each training fold. Tuning increases mean AP to 0.2211, whereas weighting reduces it slightly to 0.2206.

6.2 Supervised Evaluation

6.2.1 Evaluation Metrics

Repeat purchase is relatively uncommon (14.1% in the training data), so accuracy alone would be misleading. Our primary metric is Average Precision (AP), which summarizes precision across recall levels and is suitable for imbalanced classification.

We also use ROC-AUC for ranking quality, Precision@Top 10% to measure targeting performance among the highest-scored users, and training time for efficiency. For the final model, we report Precision, Recall, F1, Brier Score, and the confusion matrix to examine classification errors and predicted probabilities.

6.2.2 Overall Results and Cross-Validation

We compare the best configuration from each family using the mean and standard deviation (SD) across five training folds.

Table B2. Best model from each family (5-fold CV)

Model	AP (mean ± SD)	ROC-AUC (mean ± SD)	Prec.@Top 10%
GBT-C	**0.2192 ± 0.0070**	**0.6407 ± 0.0056**	**0.2611**
Logistic-B	0.2057 ± 0.0065	0.6182 ± 0.0074	0.2514
Random Forest-B	0.1993 ± 0.0078	0.6114 ± 0.0078	0.2367
Constant baseline	0.1406 ± 0.0000	0.5000 ± 0.0000	0.1406

GBT-C has the highest mean AP (0.2192). All three model families outperform the constant baseline. Adding cluster labels improves GBT AP slightly from 0.2184 (Set B) to 0.2192 (Set C), but does not improve Logistic Regression or Random Forest. However, K-Means was fitted on the outer training set before supervised cross-validation, so the CV scores for Set C may be slightly optimistic. The small AP difference should not be interpreted as a reliable improvement.

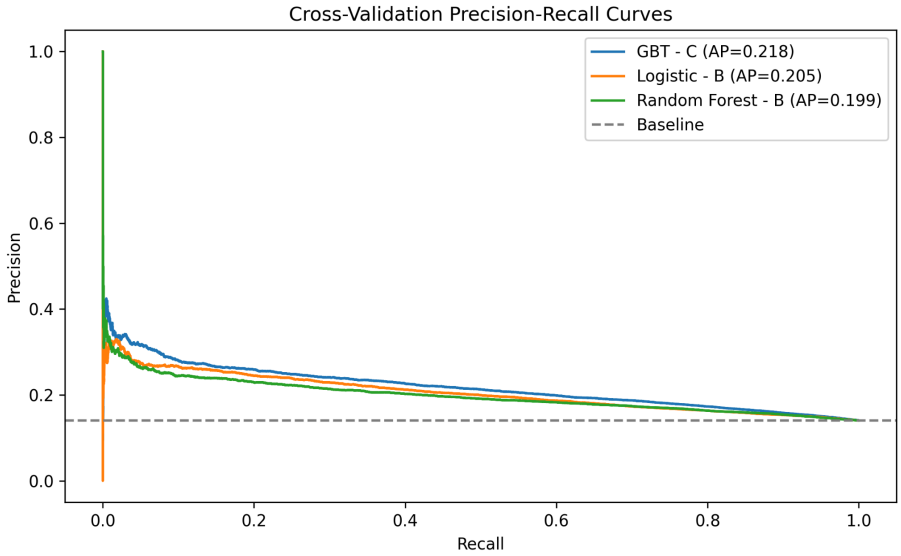


Figure B1. Out-of-fold Precision–Recall curves for the best model from each family.

Figure B1 shows that GBT-C generally maintains the highest precision over recall, although all models lose precision as recall increases.

Table B3. GBT improvement experiments (5-fold CV)

Model	**AP (mean ± SD)**
GBT-C	0.2192 ± 0.0070
GBT-D	0.2198 ± 0.0055
Tuned GBT-D	**0.2211 ± 0.0074**
Tuned GBT-D + weighting	0.2206 ± 0.0078

Engineering additional features and tuning provide only modest gains. Class weighting does not improve AP. We select unweighted Tuned GBT-D for final evaluation.

Table B4. Final performance on the temporal test set (18,292 users)

Metric	**Score**
Average Precision	0.1878
ROC-AUC	0.6312
Precision@Top 10%	0.2328
Precision (threshold 0.15)	0.1850
Recall (threshold 0.15)	0.4932
F1 (threshold 0.15)	0.2691
Brier Score	0.1062

Test AP (0.1878) is lower than five-fold training CV AP (0.2211). The test repeat-purchase rate (12.4%) is also below the training rate (14.1%), which may partly explain the AP decline. Precision@Top 10% reaches 23.3%, compared with 12.4% repeat purchases in the test set.

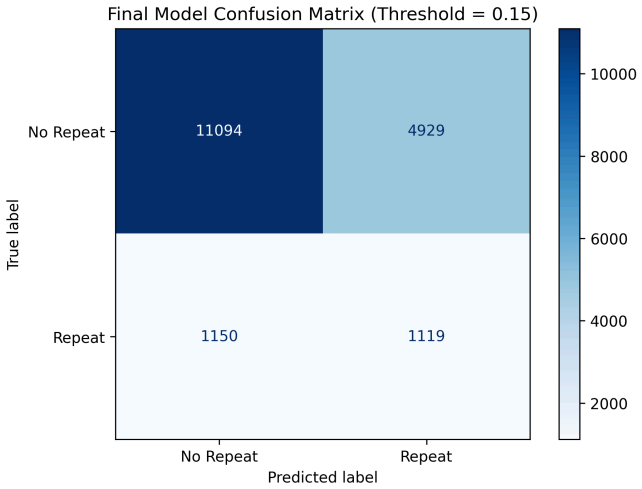


Figure B2. Held-out temporal test confusion matrix at classification threshold 0.15.

At threshold 0.15, the model correctly identifies 1,119 repeat purchasers and misses 1,150. It also generates 4,929 false positives. This illustrates the challenge of improving recall without contacting many non-repeat customers.

6.2.3 Feature Importance and Ablation Analysis

We use permutation importance on held-out CV folds to measure the change in AP when each feature is shuffled. We also remove groups of features, retrain, and compare five-fold CV AP.

Duration and active days before purchase are the strongest predictors. The cluster_label has approximately zero permutation importance, suggesting that segments add little information beyond the behavioral predictors.

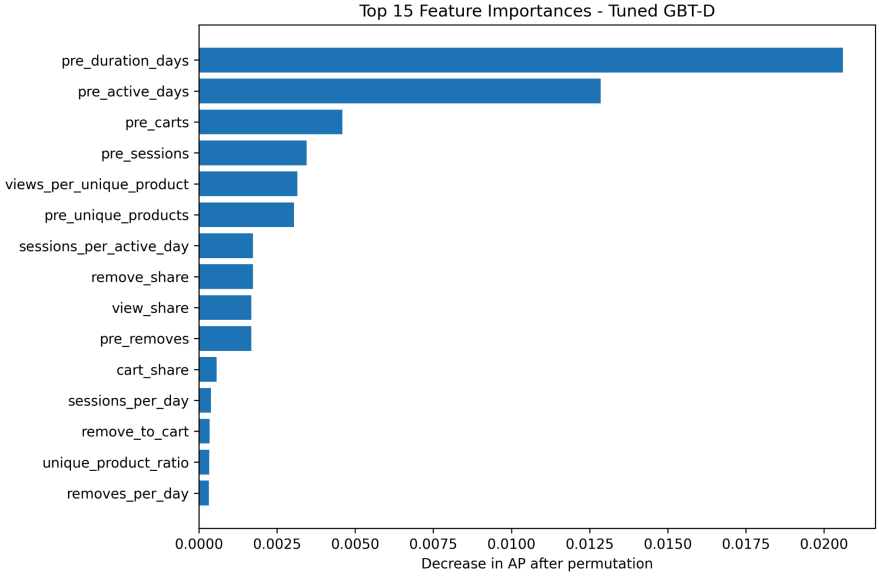


Figure B3. Top 15 permutation importances for Tuned GBT-D (mean decrease in AP).

Table B5. Feature group ablation (5-fold CV)

Removed group	Remaining features	Mean AP	AP change
None (full model)	26	0.2211	0.0000
Activity	22	0.2152	−0.0060

Removed group	Remaining features	Mean AP	AP change
Cart	17	0.2210	−0.0001
Engineered	12	0.2206	−0.0005
Cluster	25	0.2212	~0.0000

Removing the Activity group causes the largest decrease in AP (−0.0060). Removing Cart, Engineered, or Cluster features has little effect. Both analyses indicate that pre-purchase activity patterns matter most; some extra features may be unnecessary. The ablation groups overlap, so their effects should not be added together.

6.2.4 Sensitivity Analysis

We examine how tree depth affects performance while keeping the other selected GBT settings fixed. We use a three-fold CV on the training set.

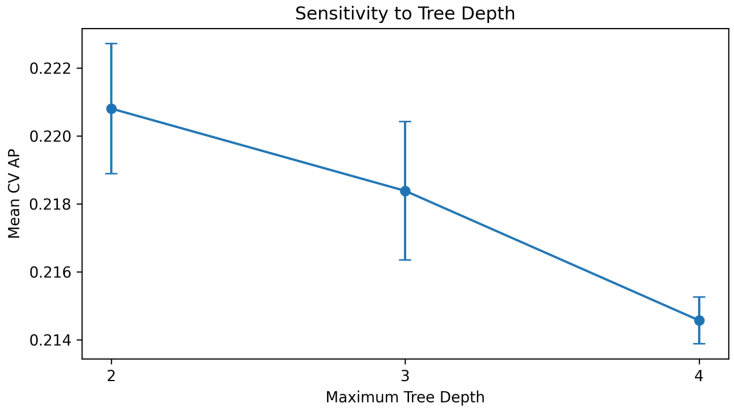


Figure B4. Change in mean cross-validation AP as maximum tree depth increases.

Depth 2 has the highest mean AP (0.2208), while depths 3 and 4 perform worse. More complex trees do not help in this experiment and may overfit. This result supports choosing a shallow tree.

6.2.5 Tradeoffs

We examine Precision, Recall, and F1 at different probability thresholds using out-of-fold predictions from three-fold CV on the training set. Lower thresholds classify more users as likely to repeat purchase.

Table B6. Precision–Recall tradeoff (training out-of-fold predictions)

Threshold	Precision	Recall	F1
0.10	0.1693	0.8321	0.2814
0.15	**0.2067**	**0.5365**	**0.2984**
0.20	0.2464	0.2964	0.2691
0.30	0.3421	0.0392	0.0703

As the threshold increases from 0.10 to 0.30, precision increases from 16.9% to 34.2%, but recall falls from 83.2% to 3.9%. Among the tested thresholds, 0.15 achieves the highest training F1 (0.2984) and is used for final test evaluation.

For marketing, a low threshold can reach more potential buyers but wastes more contacts; a high threshold prioritizes likely buyers but misses others. The best decision depends on campaign budget, reach, and contact costs.

6.2.6 Failure Analysis

We examine three specific errors in the temporal test set. The first two are high-confidence errors in opposite directions; the third is a borderline classification error.

Case 1 — High-confidence false positive (User 594472006): Predicted probability 0.5181; actual outcome: no repeat purchase. This user has 874 pre-purchase events, 423 sessions, and 188 cart additions, but also 182 cart removals. High activity may not always reflect repeat-purchase intent. A useful next step would be to add purchase value and more detailed cart-abandonment patterns.

Case 2 — Low-confidence false negative (User 597447319): Predicted probability 0.0570; actual outcome: repeat purchase. This user has only 5 events, 1 session, and 1 active day. Sparse behavioral history can hide future repeat purchasers. Product categories and first-purchase details may help.

Case 3 — Borderline false positive (User 564360373): Predicted probability approximately 0.1500; actual outcome: no repeat purchase. The user has a long pre-purchase duration (70.2 days) but only 2 active days. This prediction lies almost exactly at the classification threshold, making it sensitive to small changes. More recent activity features and cost-based threshold selection could help.

7. Discussion

7.1 Unsupervised Learning: Findings, Challenges, and Future Work

One surprising finding was that K-Means produced highly consistent clusters across different random seeds and data splits, even though its silhouette score was relatively modest. This showed us that stable clustering results do not necessarily mean the customer groups are clearly separated. We also found that K-Means and GMM identified similar and general behavioral patterns, but their evaluation metrics favored different numbers of clusters. This helped us understand that there is no single metric that determines the best clustering solution.

Our main challenge was deciding how many clusters to use and which model would best represent customer behavior. Although GMM achieved lower BIC values with more components, the resulting groups had weaker silhouette scores and were harder to interpret. We faced a trade off between capturing more behavioral variations and achieving clearer separation and simplicity. Since our goal was to identify meaningful customer segments that could also be used in supervised learning, we prioritized clearer and more interpretable groups over a more complex model. We therefore selected K-Means with k=2, while recognizing that this choice might overlook some behavioral details.

With more time and resources, we would like to explore whether our customer segments could support more targeted marketing strategies. For example, customers who spend more time browsing might benefit from personalized product recommendations, while customers with higher cart activity might respond better to purchase reminders or promotional offers. We could conduct A/B tests within each cluster to evaluate how customers respond to different strategies. This would help us determine whether the behavioral differences identified through clustering can translate into meaningful business value, rather than simply producing statistically distinct customer groups.

7.2 Supervised Learning: Findings, Challenges, and Future Work

We learned that pre-purchase activity helps predict repeat purchases, but high activity does not guarantee that a customer will return. Adding cluster labels provided little additional predictive value: GBT AP increased only from 0.2184 to 0.2192, and cluster_label had nearly zero permutation importance. Feature engineering and tuning also

produced small gains, while deeper trees performed worse. These results suggest that more features and greater model complexity do not always improve prediction.

Repeat purchases were uncommon, so we used AP as our main metric and compared models using five-fold cross-validation. We examined feature importance, feature removal, and classification errors to understand the models' limits. We also compared thresholds to balance precision and recall. AP fell from 0.2211 in training CV to 0.1878 on later users, showing that performance may change over time. We identified evaluation limitations, including overlapping outcome windows, cluster labels fitted outside individual CV folds, and repeated examination of the temporal holdout.

With more time, we would evaluate the full pipeline on a new, untouched period, ensuring that training outcomes end before test predictions begin and clustering is fitted within each training fold. We would explore first-purchase value, product categories, and recent activity as additional features. Finally, we would choose targeting thresholds based on campaign costs and test promotion strategies through randomized experiments.

8. Ethical Considerations

8.1 Unsupervised Learning

Fairness: Although our clustering model does not use demographic attributes or the repeat-purchase label, behavioral features may still reflect differences in customers' circumstances, preferences, or access to online shopping. If businesses use these segments to determine marketing strategies, some customers could receive fewer opportunities or less favorable offers based on their shopping behavior. To address this risk, businesses should monitor whether segment-based marketing decisions disproportionately disadvantage certain customer groups and ensure that all customers have access to essential services and fair treatment.

Overinterpretation of Clusters: Although K-Means with k=2 produced stable and interpretable segments, its modest silhouette score suggests that the groups are not clearly separated. Treating these clusters as fixed customer identities could lead to inaccurate assumptions about individual preferences and inappropriate marketing decisions. To mitigate this risk, cluster assignments should be used as general behavioral indicators rather than definitive customer labels, and they should be reassessed periodically as customer behavior changes.

Finally, the small difference in 30-day repeat-purchase rates (14.9% versus 13.4%) should not be used to conclude that one customer segment is inherently more valuable than the other.

8.2 Supervised Learning

Privacy: Although user identifiers are anonymized, browsing and purchase histories may disclose sensitive information about user preferences. The use of these models in personalized marketing should therefore be accompanied by responsible data management, proper consent and transparency.

Fairness: Our analysis is only relevant for customers to the extent that they have made an observed purchase and have followed up sufficiently. This model also can generate many false positives and behave differently for different customer segments. If predictions are used to target coupons or discounts, these differences can lead to unequal access to benefits. It is therefore important to evaluate the performance of the model across relevant customer segments and to use predictions as a basis for marketing decisions rather than as an automatic decision-maker.

9. Statement of Work

All team members contributed to the research design, project discussions, and final report.
Tingting Zhu : Data preparation, exploratory data analysis, and feature engineering.
Jason Gan : Unsupervised learning, clustering evaluation, and segment interpretation.
Xiaojun Xu: Supervised learning, model evaluation, and prediction analysis.
All members collaborated on interpreting results and reviewing the final report.

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11. Appendix

Table B7. Complete List of Features Used in Supervised Learning

No.	Feature	Definition	Feature Set
1	pre_events	Total number of pre-purchase events	A
2	pre_sessions	Number of pre-purchase sessions	A
3	pre_active_days	Number of distinct days with pre-purchase activity	A
4	pre_duration_days	Time in days between the first observed event and first purchase	A
5	pre_unique_products	Number of distinct products interacted with before purchase	B
6	pre_views	Number of pre-purchase product-view events	B
7	pre_carts	Number of pre-purchase add-to-cart events	B
8	pre_removes	Number of pre-purchase remove-from-cart events	B
9	view_share	pre_views / pre_events	B
10	cart_share	pre_carts / pre_events	B
11	remove_share	pre_removes / pre_events	B
12	cluster_label	Customer segment assigned by the fitted K-Means model (k = 2)	C
13	events_per_active_day	pre_events / pre_active_days	D
14	sessions_per_active_day	pre_sessions / pre_active_days	D
15	events_per_session	pre_events / pre_sessions	D
16	views_per_unique_product	pre_views / pre_unique_products	D
17	cart_to_view	pre_carts / pre_views	D
18	remove_to_cart	pre_removes / pre_carts	D
19	net_cart_actions	pre_carts − pre_removes	D
20	events_per_day	pre_events / pre_duration_days	D
21	sessions_per_day	pre_sessions / pre_duration_days	D
22	views_per_day	pre_views / pre_duration_days	D
23	carts_per_day	pre_carts / pre_duration_days	D
24	removes_per_day	pre_removes / pre_duration_days	D
25	active_day_ratio	pre_active_days / pre_duration_days	D
26	unique_product_ratio	pre_unique_products / pre_events	D

Table A1. Pairwise Standardized Euclidean Distances Between GMM Component Centers (k = 8)

	C0	C1	C2	C3	C4	C5	C6	C7
C0	0.00	2.06	3.21	3.03	1.43	1.25	1.29	2.27
C1	2.06	0.00	4.66	1.51	1.30	1.26	1.80	3.15
C2	3.21	4.66	0.00	5.14	4.45	3.64	2.98	1.89
C3	3.03	1.51	5.14	0.00	2.71	1.88	2.78	3.88
C4	1.43	1.30	4.45	2.71	0.00	1.56	1.72	3.06
C5	1.25	1.26	3.64	1.88	1.56	0.00	1.29	2.48
C6	1.29	1.80	2.98	2.78	1.72	1.29	0.00	1.40
C7	2.27	3.15	1.89	3.88	3.06	2.48	1.40	0.00